

PIKHTOVKA, Russian Federation

WMO: 295340

Lat: 55.983N

Lon: 82.700E

Elev: 417

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.4	(c) -28.4	(d) -38.9	(e) 0.6	(f) -35.1	(g) -32.4	(h) 0.9	(i) -27.3	(j) 31.8	(k) 10.4	(l) 27.7	(m) 10.4	(n) 3.3	(o) 220	(p) 0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	81.4	68.3	79.0	66.9	76.2	65.5	71.8	79.0	69.5	76.3	67.3	74.0	6.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	109.2	76.4	66.8	100.0	74.3	64.6	92.5	71.7	36.0	79.2	33.9	76.4	32.1	74.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.8	18.3	-42.0	85.6	8.3	3.0	-48.0	87.7	-52.8	89.5	-57.5	91.1	-63.5	93.3		
			-41.8	76.1	7.8	3.8	-47.5	78.8	-52.0	81.0	-56.4	83.2	-62.1	85.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	1.7	2.2	15.6	33.0	48.3	58.0	64.5	59.1	47.6	34.9	15.8	4.9
	DBStd	24.84	13.89	12.85	11.21	10.72	8.98	7.82	5.49	5.50	6.83	8.26	13.66	16.51
	HDD50	7605	1496	1337	1065	516	145	22	1	5	125	470	1026	1397
	HDD65	12024	1961	1757	1530	961	520	230	76	196	522	933	1476	1862
	CDD50	1147	0	0	0	4	92	261	449	286	53	2	0	0
	CDD65	94	0	0	0	0	2	20	60	12	0	0	0	0
	CDH74	1041	0	0	0	1	73	248	540	160	19	0	0	0
	CDH80	146	0	0	0	0	6	38	90	12	0	0	0	0
Wind	WSAvg	7.8	8.6	8.9	8.0	9.0	8.5	6.3	5.0	5.2	6.4	8.2	9.6	9.4
Precipitation	PrecAvg	18.2	0.9	0.7	0.8	1.0	1.3	2.2	2.7	2.2	1.6	1.7	1.5	1.4
	PrecMax	25.7	1.8	1.5	1.7	2.0	3.0	4.7	4.5	5.4	3.7	3.1	2.7	2.3
	PrecMin	12.8	0.2	0.0	0.1	0.2	0.1	0.7	0.5	0.4	0.6	0.6	0.9	0.6
	PrecStd	3.2	0.4	0.4	0.4	0.5	0.8	1.0	1.1	1.0	0.7	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.6	32.8	42.2	67.9	80.5	84.4	87.1	81.2	76.9	60.7	39.7	34.8
		MCWB	29.4	31.1	36.9	53.1	59.3	71.4	72.8	67.4	61.0	51.0	35.0	33.1
	2%	DB	27.6	29.6	37.6	60.0	75.5	80.3	82.2	78.4	70.1	54.8	35.6	31.5
		MCWB	26.3	27.6	32.7	48.2	57.4	67.7	69.9	66.5	60.1	47.3	32.6	30.3
	5%	DB	24.8	25.9	34.5	53.5	70.7	77.0	79.7	74.9	64.9	50.5	33.7	28.1
		MCWB	23.6	24.0	31.1	43.3	56.0	67.5	69.0	65.5	55.4	44.4	32.1	27.0
	10%	DB	21.2	22.0	31.9	47.7	66.0	73.5	77.0	71.9	60.4	46.8	31.8	25.4
		MCWB	19.9	20.6	28.8	40.1	54.2	63.4	67.4	63.9	53.2	42.5	30.4	24.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.7	31.3	38.2	55.4	63.8	75.3	75.5	72.4	63.7	52.1	36.1	33.8
		MCDB	30.2	32.2	41.4	66.1	74.0	78.4	81.3	78.4	72.3	58.6	37.8	34.9
	2%	WB	26.4	27.6	33.8	48.9	60.1	70.7	72.6	69.1	60.3	48.6	33.8	30.5
		MCDB	27.5	28.7	36.5	58.2	71.8	78.2	80.3	74.3	70.5	53.3	35.0	31.6
	5%	WB	23.6	24.0	31.7	44.1	57.8	68.0	70.5	66.5	57.0	45.6	32.1	27.3
		MCDB	24.8	26.0	34.0	51.0	68.2	75.2	77.8	73.6	63.2	49.9	33.4	28.2
	10%	WB	19.7	20.6	29.0	41.0	54.8	65.0	68.4	64.3	54.0	42.3	30.3	24.0
		MCDB	21.0	22.1	31.5	47.4	64.8	71.2	75.4	70.5	59.4	46.3	31.9	25.2
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.9	21.2	17.2	20.6	19.7	19.4	20.6	19.0	13.5	13.2	14.9
		MCDBR	12.9	15.9	19.6	26.0	31.0	25.5	24.7	26.0	27.8	22.5	12.8	15.5
		MCWBR	11.0	14.4	16.4	16.2	17.4	16.4	14.1	15.5	17.5	14.9	12.0	14.7
	5% WB	MCDBR	11.6	14.7	15.4	24.6	26.7	23.1	22.8	22.7	26.3	20.6	12.7	15.5
		MCWBR	11.5	13.7	13.6	16.6	17.1	17.1	15.2	15.8	18.2	14.8	12.2	15.3
Clear-Sky Solar Irradiance	taub		0.289	0.323	0.372	0.405	0.405	0.408	0.439	0.409	0.369	0.351	0.318	0.291
	taud		2.005	1.974	1.915	2.069	2.220	2.289	2.200	2.274	2.334	2.244	2.062	1.955
	Ebn at Noon		180	222	240	255	265	265	253	253	247	216	167	141
	Edh at Noon		21	33	45	45	41	39	42	36	30	25	20	17
All-Sky Solar Radiation	RadAvg		267	587	1054	1435	1651	1857	1674	1375	917	462	221	163
	RadStd		25	35	65	109	166	187	119	107	117	48	39	27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page